

LVM Storage Management in OpenSUSE

Problem Statement

The existing OpenSUSE virtual machine does not have sufficient storage space to meet application and data storage requirements. To resolve this, additional virtual disks must be attached and configured using Linux Logical Volume Manager (LVM). The storage should be organized into a Volume Group and divided into separate Logical Volumes, which are then formatted, mounted, and configured for persistent access

Objective

To configure and manage storage in an OpenSUSE virtual machine using Logical Volume Manager (LVM) by adding new virtual disks, creating a Volume Group, provisioning Logical Volumes, and mounting them persistently for application and data storage.

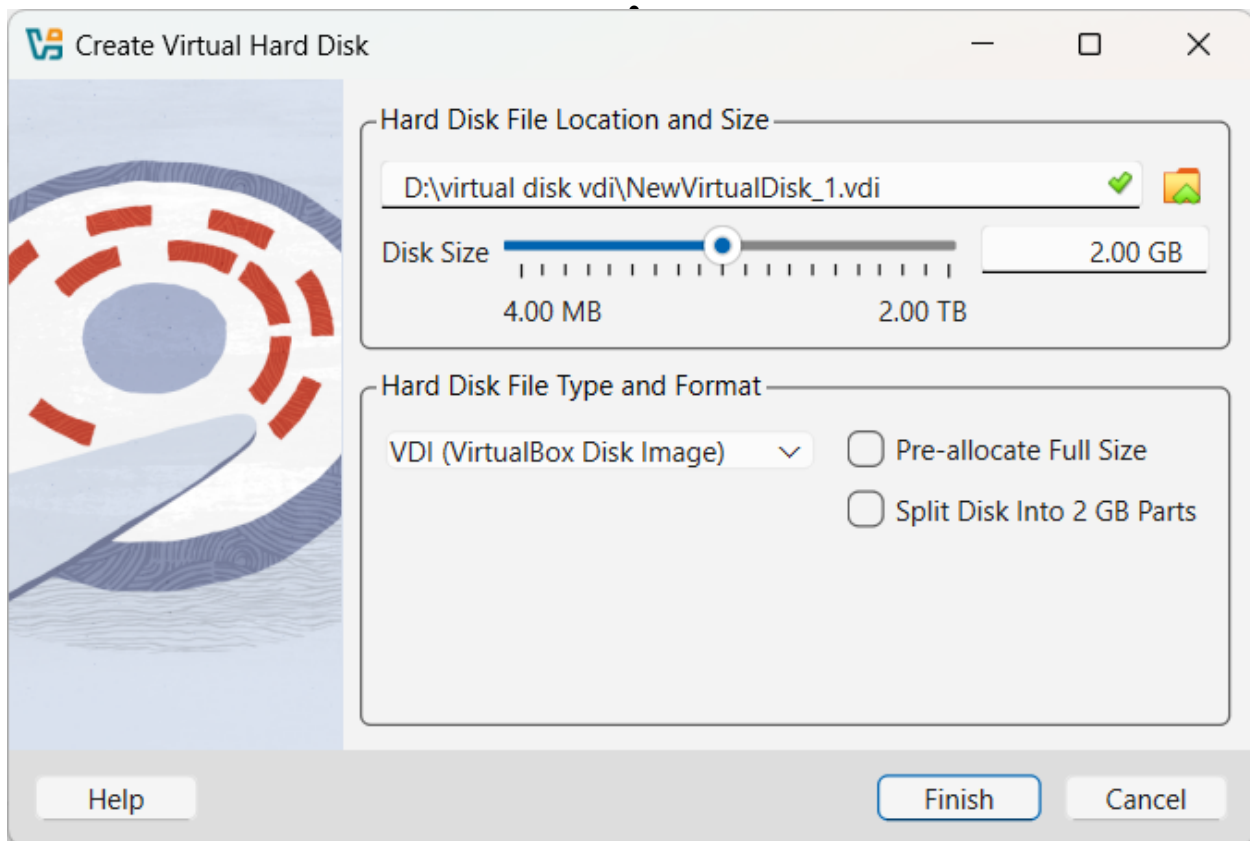
Key Requirements

- Two additional virtual disks of **2 GB** and **3 GB** attached to the VM.
- Create a Physical Volume (PV)
- Create a Volume Group
- Create two Logical Volumes
- Format both Logical Volumes with a Linux filesystem (e.g., ext4).
- Create mount directories:
 - /mnt/app1_data
 - /mnt/test_data
- Mount the Logical Volumes to their respective directories.
- Configure **persistent mounting** using the /etc/fstab file.

Implementation Steps

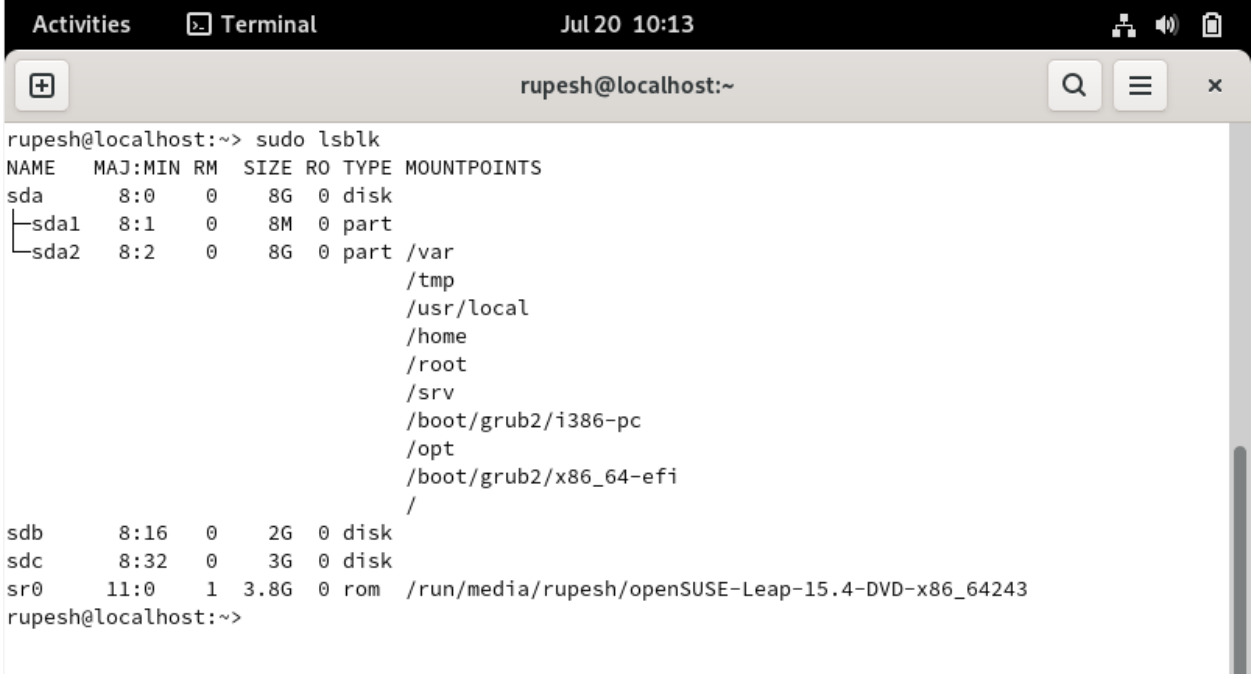
Step 1: Add New Virtual Disks

- Open VirtualBox Settings.
- Attach two new virtual hard disks:
 - Disk 1 – **2 GB**
 - Disk 2 – **3 GB**



Step 2: Verify the Newly Added Disks

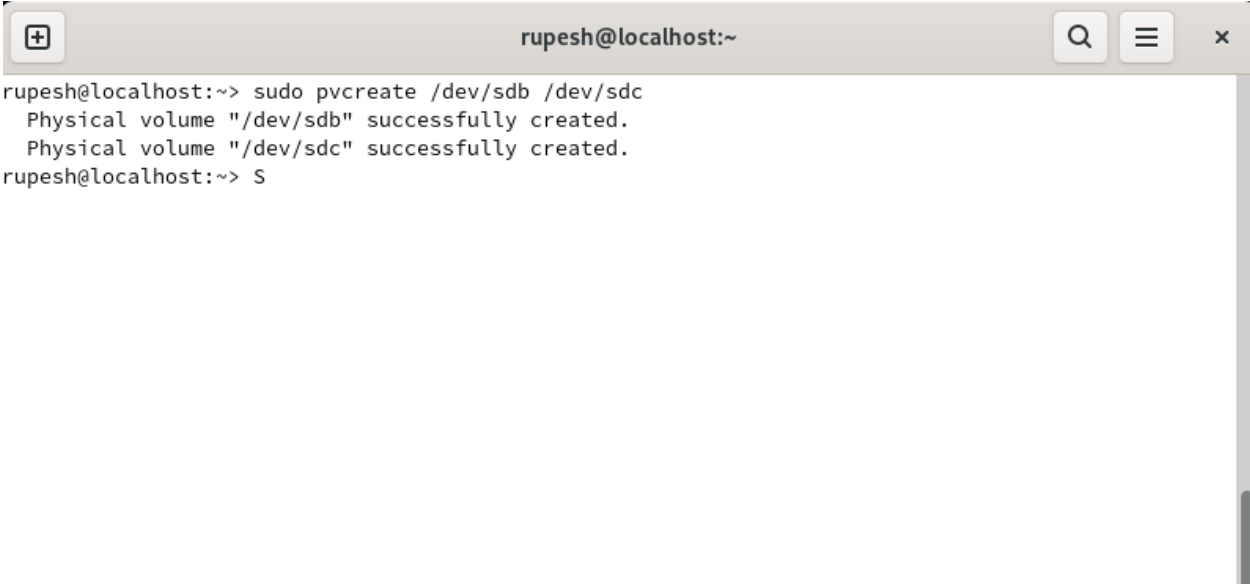
- List available storage devices.
- Identify the newly attached disks.



```
Activities Terminal Jul 20 10:13
rupesh@localhost:~
rupesh@localhost:~> sudo lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0   8G  0 disk
├─sda1       8:1    0    8M  0 part
└─sda2       8:2    0   8G  0 part /var
                        /tmp
                        /usr/local
                        /home
                        /root
                        /srv
                        /boot/grub2/i386-pc
                        /opt
                        /boot/grub2/x86_64-efi
                        /
sdb          8:16    0   2G  0 disk
sdc          8:32    0   3G  0 disk
sr0         11:0    1  3.8G  0 rom  /run/media/rupesh/openSUSE-Leap-15.4-DVD-x86_64243
rupesh@localhost:~>
```

Step 3 :- Create Physical Volumes (PVs) :-

`pvcreate /dev/sdb /dev/sdc`



```
rupesh@localhost:~> sudo pvcreate /dev/sdb /dev/sdc
Physical volume "/dev/sdb" successfully created.
Physical volume "/dev/sdc" successfully created.
rupesh@localhost:~> S
```

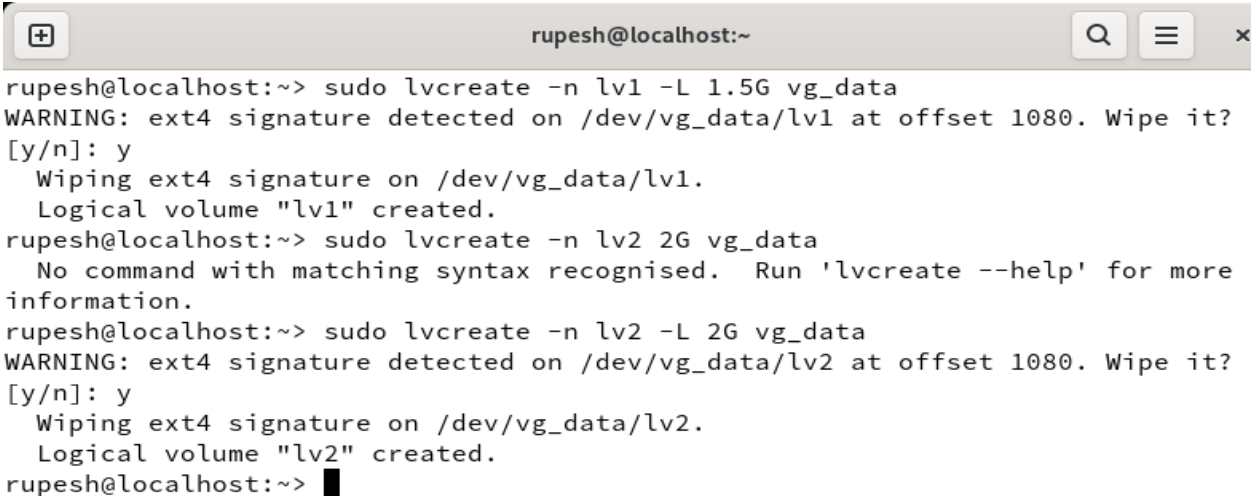
Step 4 :- Create the Volume Group

- **Vgcreate vg_data /dev/sdb /dev/sdc**
- Combine both Physical Volumes into a Volume Group named vg_data.
- Verify that the Volume Group has approximately 5 GB of available storage.

```
rupesh@localhost:~> sudo vgcreate vg_data /dev/sdb /dev/sdc
Volume group "vg_data" successfully created
rupesh@localhost:~>
```

Step 5 :- Create Logical Volumes.

- **Lvcreate -n lv1 -L 1.5G vg_data**
- Create lv1 with a size of 1.5 GB.
- Create lv2 with a size of 2 GB.
- Verify both Logical Volumes.



```
rupesh@localhost:~> sudo lvcreate -n lv1 -L 1.5G vg_data
WARNING: ext4 signature detected on /dev/vg_data/lv1 at offset 1080. Wipe it?
[y/n]: y
Wiping ext4 signature on /dev/vg_data/lv1.
Logical volume "lv1" created.
rupesh@localhost:~> sudo lvcreate -n lv2 2G vg_data
No command with matching syntax recognised. Run 'lvcreate --help' for more
information.
rupesh@localhost:~> sudo lvcreate -n lv2 -L 2G vg_data
WARNING: ext4 signature detected on /dev/vg_data/lv2 at offset 1080. Wipe it?
[y/n]: y
Wiping ext4 signature on /dev/vg_data/lv2.
Logical volume "lv2" created.
rupesh@localhost:~>
```

Step 6: Create Filesystems

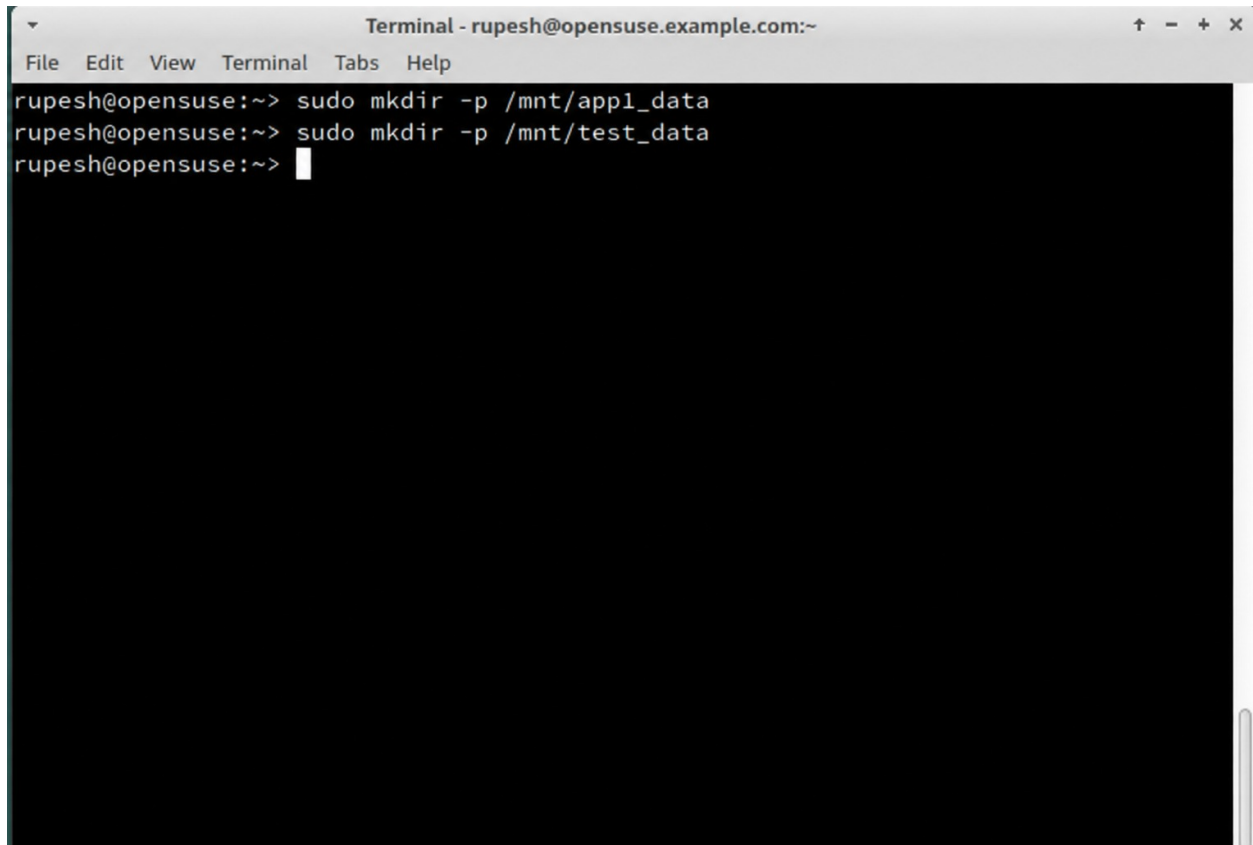
- Format both Logical Volumes using the ext4 filesystem (or another supported Linux filesystem).

```
rupesh@localhost:~> sudo mkfs.ext4 /dev/vg_data/lv1
mke2fs 1.46.4 (18-Aug-2021)
Creating filesystem with 393216 4k blocks and 98304 inodes
Filesystem UUID: 9b94d112-12de-4314-b2df-faa493fc234a
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done
```

Step 7 :- Create Mount Directories

- **/mnt/app1_data**
- **/mnt/test_data**

A terminal window titled "Terminal - rupesh@opensuse.example.com:~" with a menu bar (File, Edit, View, Terminal, Tabs, Help). The terminal shows the following commands and output:

```
rupesh@opensuse:~> sudo mkdir -p /mnt/app1_data
rupesh@opensuse:~> sudo mkdir -p /mnt/test_data
rupesh@opensuse:~> 
```

Step 8 :- Configure Persistent Mounting

- Obtain the UUIDs of both Logical Volumes.
- Add the appropriate entries to the /etc/fstab file.
- Test the configuration to ensure there are no errors.

```
Terminal - rupesh@opensuse.example.com:~
File Edit View Terminal Tabs Help
UUID=e96e7a65-0796-42a2-989c-873e44ecb8e8 /usr/local btrfs subvol=@/usr/local
/usr/local 0 0
UUID=e96e7a65-0796-42a2-989c-873e44ecb8e8 /tmp btrfs subvol=@/tmp
/tmp 0 0
UUID=e96e7a65-0796-42a2-989c-873e44ecb8e8 /srv btrfs subvol=@/srv
/srv 0 0
UUID=e96e7a65-0796-42a2-989c-873e44ecb8e8 /root btrfs subvol=@/root
/root 0 0
UUID=e96e7a65-0796-42a2-989c-873e44ecb8e8 /opt btrfs subvol=@/opt
/opt 0 0
UUID=e96e7a65-0796-42a2-989c-873e44ecb8e8 /home btrfs subvol=@/home
/home 0 0
UUID=e96e7a65-0796-42a2-989c-873e44ecb8e8 /boot/grub2/x86_64-efi btrfs subvol=@/boot/grub2/x86_64-efi
/boot/grub2/x86_64-efi 0 0
UUID=e96e7a65-0796-42a2-989c-873e44ecb8e8 /boot/grub2/i386-pc btrfs subvol=@/boot/grub2/i386-pc
/boot/grub2/i386-pc 0 0
UUID=e96e7a65-0796-42a2-989c-873e44ecb8e8 /.snapshots btrfs subvol=@/.snapshots
/.snapshots 0 0
UUID=87967ef7-aecf-44b5-bd36-d689cf044b81 swap swap defaults
0 0
UUID=5c67f9da-d891-413b-d67d-d6c0d5cc32e5 /mnt/app1_data ext4 defaults
0 0
UUID=3ec1e0e7-348a-442d-bf7d-bace4950370c /mnt/test_data ext4 defaults
0 0
"/etc/fstab" 14L, 1500B 14,97-108 Bot
```

Step 9 :- Verify the Configuration

- Confirm both Logical Volumes are mounted correctly.
- Check available disk space.

```
rupesh@opensuse:~> df -hT
Filesystem      Type      Size  Used Avail Use% Mounted on
devtmpfs        devtmpfs  4.0M   0    4.0M  0% /dev
tmpfs           tmpfs     2.0G   4.0K  2.0G   1% /dev/shm
tmpfs           tmpfs     782M   26M   756M   4% /run
tmpfs           tmpfs     4.0M   0    4.0M   0% /sys/fs/cgroup
/dev/sda2        btrfs     23G   5.7G   17G   26% /
/dev/sda2        btrfs     23G   5.7G   17G   26% /.snapshots
/dev/sda2        btrfs     23G   5.7G   17G   26% /boot/grub2/i386-pc
/dev/sda2        btrfs     23G   5.7G   17G   26% /boot/grub2/x86_64-efi
/dev/sda2        btrfs     23G   5.7G   17G   26% /home
/dev/sda2        btrfs     23G   5.7G   17G   26% /root
/dev/sda2        btrfs     23G   5.7G   17G   26% /opt
/dev/sda2        btrfs     23G   5.7G   17G   26% /usr/local
/dev/sda2        btrfs     23G   5.7G   17G   26% /srv
/dev/sda2        btrfs     23G   5.7G   17G   26% /tmp
/dev/sda2        btrfs     23G   5.7G   17G   26% /var
tmpfs           tmpfs     391M   76K   391M   1% /run/user/1000
/dev/mapper/vg_data-lv1 ext4      1.5G   24K   1.4G   1% /mnt/appl_data
/dev/mapper/vg_data-lv2 ext4      2.0G   24K   1.8G   1% /mnt/test_data
rupesh@opensuse:~>
```

Conclusion :-

The task was completed successfully by configuring Logical Volume Management (LVM) on the OpenSUSE virtual machine. Two additional disks were added and combined into a Volume Group named `vg_data`. Two Logical Volumes, `lv1` (1.5 GB) and `lv2` (2 GB), were created, formatted with the `ext4` filesystem, and mounted to `/mnt/app1_data` and `/mnt/test_data`. Persistent mounting was configured through the `/etc/fstab` file, ensuring that both logical volumes are automatically mounted after every system reboot. The final verification confirmed that the LVM configuration and mount points were functioning correctly.